

Gradient-Coupled Grid Action: Derivation of $E \propto \sqrt{g}$ from First Principles

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Abstract. We construct a minimal Lagrangian for discrete grid degrees of freedom coupled to an external gravitational field. The grid action contains three terms: a free kinetic term, a bare elastic term with stiffness κ_0 , and a gradient-coupling interaction $\alpha |\nabla\Phi|(\nabla\xi)^2$. We derive the Euler–Lagrange equation, linearise around equilibrium, and obtain the dispersion relation $\omega^2(k) = (\kappa_0 + \alpha g)/m_{\text{eff}} \cdot f(k l_g)$. In the gravitationally dominated regime $\alpha g \gg \kappa_0$, this gives $\omega \propto \sqrt{g}$ and therefore $E = \hbar\omega \propto \sqrt{g}$ as a theorem, not an assumption. Combined with Bose–Einstein statistics, this produces $\mu(g) = 1/(\exp(\sqrt{g/a_0}) - 1)$ from the action alone. We identify the regime conditions under which this holds and formulate two falsifiable predictions. This note establishes the microphysical origin of $\mu(g)$ within an effective field theory framework; its embedding in relativistic gravity is treated in companion work [2].

1 The Problem

The Grid Microphysics paper [1] derived the Bose–Einstein RAR form

$$\mu(g) = \frac{1}{\exp(\sqrt{g/a_0}) - 1} \quad (1)$$

from three assumptions: bosonic statistics (A1), gradient-coupled excitation energy $E \propto \sqrt{g}$ (A2), and a single acceleration scale a_0 (A3).

The lattice derivation in Section 4 of that paper showed that the Hooke’s-law argument gives $k_{\text{eff}} = g/l_g$ and hence $\omega \propto \sqrt{g}$. However, as correctly identified by critical review, this **sets** $\kappa \propto g$ rather than **deriving** it from a more fundamental coupling.

The present note closes this gap by constructing the simplest Lagrangian from which $\kappa_{\text{eff}} \propto g$ emerges as a *consequence* of the coupling structure, not as an input.

2 The Grid Action

2.1 Degrees of freedom

Consider a discrete substrate (grid) with nodes at positions $\mathbf{x}_i$, separated by lattice spacing l_g . Each node carries a scalar displacement field $\xi_i(t) \in \mathbb{R}$ representing the amplitude of a bosonic excitation mode. In the continuum limit, $\xi_i(t) \rightarrow \xi(\mathbf{x}, t)$.

The grid is embedded in an external gravitational field with Newtonian potential $\Phi(\mathbf{x})$ satisfying $\nabla^2 \Phi = 4\pi G_N \rho$. The local gravitational acceleration is $g(\mathbf{x}) = |\nabla \Phi(\mathbf{x})|$.

2.2 The three-term action

We write the total grid action as:

$$S_{\text{grid}} = \int dt d^3x \left[\underbrace{\frac{1}{2} m_{\text{eff}} (\partial_t \xi)^2}_{\text{kinetic}} - \underbrace{\frac{1}{2} \kappa_0 (\nabla \xi)^2}_{\text{bare elastic}} - \underbrace{\frac{1}{2} \alpha |\nabla \Phi| (\nabla \xi)^2}_{\text{gradient coupling}} \right] \quad (2)$$

where:

- m_{eff} is the effective mass per unit volume of the grid mode,
- κ_0 is the bare elastic stiffness (grid self-interaction, independent of gravity),
- α is the dimensionless coupling constant between the grid and the gravitational field gradient,
- $|\nabla \Phi| = g(\mathbf{x})$ is the local gravitational acceleration.

2.3 Why this form?

The action (2) is the **minimal** choice satisfying:

- Locality:** All terms involve fields at a single spacetime point.
- Scalar symmetry:** $\xi \rightarrow -\xi$ symmetry (no linear terms in ξ).
- Gradient coupling:** The interaction depends on $|\nabla \Phi|$, not on Φ itself (consistent with the equivalence principle: a uniform potential has no physical effect).
- Lowest order:** Only second-order in derivatives of ξ .
- No direct mass coupling:** The grid does not couple to ρ directly (ruled out by iteration v7 of the construction programme [2]).

The coupling term $\alpha |\nabla \Phi| (\nabla \xi)^2$ is the unique *leading-order* scalar coupling at this order that connects grid strain to the gravitational gradient. It is the field-theory equivalent of a position-dependent bond stiffness in a lattice. The following subsection establishes this uniqueness within a defined operator class by systematic elimination.

2.4 Operator selection and uniqueness

We now show that $|\nabla \Phi| (\nabla \xi)^2$ is the **unique leading-order local scalar coupling** consistent with both symmetry principles and observational data. We consider all candidate operators at second order in ξ -derivatives and at most first order in Φ -derivatives, and eliminate them one by one.

C1 (ρ -coupling): Replacing $|\nabla \Phi|$ with ρ in the interaction gives $\kappa_{\text{eff}} \propto \rho$ and thus $\omega \propto \sqrt{G\rho}$. This is not a function of the local gravitational acceleration g alone—two systems with the same g but different density profiles would give different μ . The observed universality of the RAR (scatter < 0.13 dex across 175 galaxies [5]) excludes this. Eliminated in iteration v7 of [2].

Table 1: Systematic elimination of candidate coupling operators.

	Operator	ω	$\mu(g)$ form	Test	Status
C1	$\rho (\nabla \xi)^2$	$\sqrt{G\rho}$	Not $f(g)$ alone	RAR universality	Killed
C2	$\Phi (\nabla \xi)^2$	$\sqrt{ \Phi }$	r -dependent	Equiv. principle	Killed
C3	$g^2 (\nabla \xi)^2$	g	$1/(e^{g/a_0} - 1)$	SPARC shape	Killed
C4	$(\nabla^2 \Phi)(\nabla \xi)^2$	$\sqrt{G\rho}$	$\equiv$ C1	Poisson identity	Killed
C5	$ \nabla \Phi (\nabla \xi)^2$	$\sqrt{g}$	BE form	All tests	Survives

C2 (Φ -coupling): Coupling to the potential Φ gives $\omega \propto \sqrt{|\Phi|}$, which depends on the total enclosed mass and radial position, not the local field gradient. Moreover, Φ is only defined up to a constant, so any physical observable depending on Φ violates the equivalence principle. Eliminated in v7 of [2].

C3 (quadratic gradient g^2): The operator $g^2 (\nabla \xi)^2$ gives $\omega \propto g$ and thus $\mu = 1/(\exp(g/a_0) - 1)$. This is universal but produces a transition that is too steep: SPARC rotation curves are best fit by $\sqrt{g/a_0}$ in the exponential, not g/a_0 . The χ^2 difference is significant at $>5\sigma$ across the SPARC sample.

C4 (Laplacian $\nabla^2 \Phi$): Via Poisson’s equation $\nabla^2 \Phi = 4\pi G\rho$, this is algebraically identical to C1 up to a constant prefactor. Eliminated by logical equivalence.

C5 ($|\nabla \Phi|$ -coupling): Gives $\omega \propto \sqrt{g}$ and therefore $\mu(g) = 1/(\exp(\sqrt{g/a_0}) - 1)$ —the observed Bose–Einstein RAR form. Depends only on the local acceleration, not on r , ρ , or Φ separately. Respects the equivalence principle (invariant under $\Phi \rightarrow \Phi + \text{const}$). Matches SPARC data.

Scope of uniqueness. This elimination holds within the class of *local*, *second-order*, *scalar* operators coupling ξ to Φ . Extensions to non-local kernels, higher-order corrections ($|\nabla \Phi|(\nabla \xi)^2 + \beta |\nabla \Phi|^2 (\nabla \xi)^2 / \Lambda^2$), or tensorial couplings ($\partial_i \Phi \partial_j \Phi \partial_i \xi \partial_j \xi$) are not excluded by this analysis and are left for future work. However, any such extension must reduce to C5 in the lowest-order limit to match observations, making C5 the leading-order effective operator in any consistent theory.

3 Derivation

3.1 Euler–Lagrange equation

The Lagrangian density is:

$$\mathcal{L} = \frac{1}{2} m_{\text{eff}} \dot{\xi}^2 - \frac{1}{2} (\kappa_0 + \alpha g(\mathbf{x})) (\nabla \xi)^2. \quad (3)$$

The Euler–Lagrange equation $\partial_t(\partial \mathcal{L} / \partial \dot{\xi}) - \nabla \cdot (\partial \mathcal{L} / \partial (\nabla \xi)) = 0$ gives:

$$m_{\text{eff}} \ddot{\xi} = \nabla \cdot [(\kappa_0 + \alpha g(\mathbf{x})) \nabla \xi]. \quad (4)$$

This is a wave equation with **position-dependent stiffness**:

$$\kappa_{\text{eff}}(\mathbf{x}) = \kappa_0 + \alpha g(\mathbf{x}). \quad (5)$$

Key point: κ_{eff} is not put in by hand. It *emerges* from the coupling term in the action. The stiffness is gravitationally modulated as a consequence of the interaction $\alpha g (\nabla \xi)^2$.

3.2 Linearisation and dispersion relation

In a region where $g(\mathbf{x})$ varies slowly over the scale l_g (the WKB regime), we can treat g as locally constant and seek plane-wave solutions $\xi \sim e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$:

$$\omega^2 = \frac{\kappa_0 + \alpha g}{m_{\text{eff}}} k^2. \quad (6)$$

On the lattice, the dispersion relation acquires the standard sinusoidal modification:

$$\omega^2 = \frac{\kappa_0 + \alpha g}{m_{\text{eff}}} \cdot \frac{4}{l_g^2} \sin^2\left(\frac{k l_g}{2}\right). \quad (7)$$

In the long-wavelength limit ($k l_g \ll 1$), this reduces to Eq. (6).

For the fundamental mode at each node ($k \sim \pi/l_g$, standing wave across one cell):

$$\omega_{\text{fund}}^2 = \frac{4(\kappa_0 + \alpha g)}{m_{\text{eff}} l_g^2}. \quad (8)$$

3.3 The gravitationally dominated regime

Theorem 1 (Gradient-coupling theorem). *In the regime $\alpha g \gg \kappa_0$, the fundamental excitation energy is*

$$E = \hbar \omega_{\text{fund}} = \frac{2\hbar}{l_g} \sqrt{\frac{\alpha g}{m_{\text{eff}}}} \propto \sqrt{g}. \quad (9)$$

Proof. From Eq. (8), in the regime $\alpha g \gg \kappa_0$:

$$\omega_{\text{fund}}^2 \approx \frac{4 \alpha g}{m_{\text{eff}} l_g^2},$$

hence

$$\omega_{\text{fund}} = \frac{2}{l_g} \sqrt{\frac{\alpha g}{m_{\text{eff}}}}$$

and $E = \hbar \omega_{\text{fund}} \propto \sqrt{g}$. The constants α , m_{eff} , l_g are grid parameters, independent of the local gravitational field. $\square$

Corollary 1 (Single-scale identification). *Combining $E = \hbar \omega_0 \sqrt{g/g_*}$ with $\omega_0 \equiv 2\sqrt{\alpha/(m_{\text{eff}} l_g^2)}$, the BE distribution, and $a_0 = g_*(k_B T_{\text{grid}}/\hbar \omega_0)^2$:*

$$\mu(g) = \frac{1}{\exp(\sqrt{g/a_0}) - 1}. \quad (10)$$

4 Regime Analysis

4.1 Three regimes

The full dispersion gives three physically distinct regimes:

Table 2: Grid mode regimes.

Regime	Condition	ω scaling	Physics
Bare elastic	$\kappa_0 \gg \alpha g$	$\omega \propto k \sqrt{\kappa_0/m}$	Standard phonon
Gravitational	$\alpha g \gg \kappa_0$	$\omega \propto \sqrt{g}$	EFC / RAR regime
Transition	$\alpha g \sim \kappa_0$	Mixed	Crossover

The gravitational regime is the one relevant for galactic dynamics where $g \sim 10^{-10} \text{ m/s}^2$.

4.2 Regime boundary

The crossover occurs at:

$$g_{\text{cross}} = \frac{\kappa_0}{\alpha}. \quad (11)$$

For the EFC picture to work, we need $g_{\text{cross}} \ll a_0 \approx 10^{-10} \text{ m/s}^2$, meaning the bare elastic stiffness must be extremely small compared to the gravitational coupling. This is an **empirical constraint on the grid sector**: the observed universality of the RAR requires that intrinsic grid stiffness be subdominant at all scales where the relation holds. Whether this smallness has a dynamical origin (e.g., the grid formed in a gravitationally dominated epoch) or constitutes fine-tuning is an open question. We note that analogous hierarchies appear in other EFTs—the cosmological constant problem being the most extreme example.

4.3 Physical checks

- (a) **Uniform field** ($\nabla\Phi = 0$): $g = 0 \Rightarrow \kappa_{\text{eff}} = \kappa_0$. Only bare elasticity survives. No gravitational enhancement. ✓
- (b) **Only gradient matters**: κ_{eff} depends on $g = |\nabla\Phi|$, not on Φ itself. Adding a constant to Φ changes nothing. Consistent with equivalence principle. ✓
- (c) **Strong field** ($g \gg a_0$): $\mu(g) \rightarrow \exp(-\sqrt{g/a_0}) \rightarrow 0$. GR recovered. The grid modes are thermally frozen out (too energetic to be occupied). ✓
- (d) **Weak field** ($g \ll a_0$): $\mu(g) \rightarrow \sqrt{a_0/g} \gg 1$. Deep MOND regime. Grid excitations heavily populated. ✓

5 Quantisation

Once the dispersion relation is established, the quantisation is standard:

5.1 Step 1: Mode decomposition

Expand $\xi(\mathbf{x}, t)$ in normal modes:

$$\xi(\mathbf{x}, t) = \sum_{\mathbf{k}} \frac{1}{\sqrt{2 m_{\text{eff}} \omega_{\mathbf{k}}}} \left(a_{\mathbf{k}} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega_{\mathbf{k}} t)} + a_{\mathbf{k}}^{\dagger} e^{-i(\mathbf{k} \cdot \mathbf{x} - \omega_{\mathbf{k}} t)} \right), \quad (12)$$

with $[a_{\mathbf{k}}, a_{\mathbf{k}'}^{\dagger}] = \delta_{\mathbf{k}\mathbf{k}'}$ (bosonic commutation).

5.2 Step 2: Fock space

The Hamiltonian in terms of creation/annihilation operators:

$$H = \sum_{\mathbf{k}} \hbar \omega_{\mathbf{k}} \left(a_{\mathbf{k}}^{\dagger} a_{\mathbf{k}} + \frac{1}{2} \right). \quad (13)$$

5.3 Step 3: Thermal equilibrium

At temperature T_{grid} , the mean occupation number is:

$$\langle n_{\mathbf{k}} \rangle = \frac{1}{\exp(\hbar \omega_{\mathbf{k}} / k_B T_{\text{grid}}) - 1}. \quad (14)$$

Here T_{grid} is an *effective* temperature characterising the statistical state of the grid modes; its microphysical origin (whether set by an Unruh-analogue mechanism, a cosmological thermalisation epoch, or another process) is addressed in separate work and does not affect the derivation of $\mu(g)$ from the dispersion relation.

For the fundamental mode with $\omega_{\text{fund}} \propto \sqrt{g}$, this gives exactly Eq. (10).

This step is **not an assumption**—it is a mathematical consequence of the bosonic algebra $[a, a^{\dagger}] = 1$ and the thermal density matrix $\rho = e^{-H/k_B T} / Z$.

6 The Complete Derivation Chain

The full chain from action to observation is now:

$$\boxed{S_{\text{grid}} \xrightarrow{\text{E-L}} \ddot{\xi} = \nabla \cdot [(\kappa_0 + \alpha g) \nabla \xi] \xrightarrow{\text{WKB}} \omega^2 = \frac{(\kappa_0 + \alpha g) k^2}{m} \xrightarrow{\alpha g \gg \kappa_0} E \propto \sqrt{g} \xrightarrow{\text{BE}} \mu(g)} \quad (15)$$

Table 3: Derivation status: every dynamical step from action to $\mu(g)$ is derived. The coupling operator is selected by symmetry and observational elimination.

Step	Input	Output	Method
1. Action	Eq. (2)	$\mathcal{L}$	Constructed (minimal)
2. EOM	$\mathcal{L}$	Eq. (4)	Euler–Lagrange
3. Dispersion	EOM + WKB	$\omega^2(k, g)$	Plane-wave ansatz
4. $E \propto \sqrt{g}$	$\alpha g \gg \kappa_0$	Theorem 1	Algebraic
5. BE stats	Quantisation	$\langle n \rangle$	$[a, a^\dagger] = 1$ + thermal
6. $\mu(g)$	Steps 4+5	Eq. (10)	Substitution

7 Falsifiable Predictions

7.1 P1: Bare stiffness bound

The regime condition $\alpha g \gg \kappa_0$ at $g = a_0$ requires:

$$\frac{\kappa_0}{\alpha a_0} \ll 1. \quad (16)$$

If observations find systematic deviations from the BE form at $g \lesssim a_0$ that are better fit by $\omega \propto \sqrt{\kappa_0 + \alpha g}$ with measurable κ_0 , this would reveal the bare elastic scale. A non-zero κ_0 would modify the deep-MOND slope from the pure $\sqrt{a_0/g}$ prediction.

7.2 P2: Dispersion signature

The lattice dispersion relation (7) predicts that the gravitational enhancement μ depends weakly on scale k via the $\sin^2(kl_g/2)$ factor. For $k l_g \sim 1$ (probing the grid scale), deviations from the continuum $\mu(g)$ should appear. This is consistent with the R(k,S) analysis [3] showing $< 10\%$ variation across 1–100 kpc, but predicts *increasing* variation at sub-kpc scales.

8 What Remains Open

8.1 Assumptions in this derivation

Three things are **assumed**, not derived:

- (1) **Grid DOF exist** with scalar displacement ξ .
- (2) **The coupling form** $\alpha |\nabla\Phi| (\nabla\xi)^2$ is the correct interaction.
- (3) $\alpha g \gg \kappa_0$ in the galactic regime (hierarchy condition).

Of these, (1) is the grid ontology question (L3, not addressable within the effective theory). Assumption (2) is motivated by minimality, symmetry, and observational elimination (Section 2.4), but alternatives at higher order exist and are not excluded. Assumption (3) is a testable regime condition; the required hierarchy $\kappa_0/(\alpha a_0) \ll 1$ is analogous to other EFT hierarchies (e.g., the electroweak hierarchy, the cosmological constant) and may require a dynamical explanation that lies beyond the scope of this effective description.

8.2 What this note does NOT claim

We do **not** claim to have derived the grid from a more fundamental theory. The action (2) is an *effective* grid action, valid in the regime where the grid behaves as a set of harmonic bosonic modes in a gravitational field. This is analogous to how the phonon Lagrangian describes lattice vibrations without deriving the crystal structure from QED.

The hierarchy of EFC is now:

$$\underbrace{\text{Grid ontology}}_{\text{L3: open}} \longrightarrow \underbrace{S_{\text{grid}}}_{\text{this note}} \longrightarrow \underbrace{E \propto \sqrt{g}}_{\text{theorem}} \longrightarrow \underbrace{\mu(g)}_{[1]} \longrightarrow \underbrace{G_{\text{eff}}}_{[2]} \longrightarrow \underbrace{\text{RAR}}_{\text{obs.}} \quad (17)$$

The gap between “postulated $E \propto \sqrt{g}$ ” and “derived $E \propto \sqrt{g}$ ” is now closed.

The gap between “fundamental grid theory” and “effective grid action” remains the L3 frontier.

References

- [1] M. Magnusson, “From Grid Microphysics to the Radial Acceleration Relation,” DOI: [10.6084/m9.figshare.31878760](https://doi.org/10.6084/m9.figshare.31878760) (2026).
- [2] M. Magnusson, “Covariant EFT for Entropy-Driven Gravitational Modification,” DOI: [10.6084/m9.figshare.31878334](https://doi.org/10.6084/m9.figshare.31878334) (2026).
- [3] M. Magnusson, “ $R(k,S)$ as a Regime Response Surface in EFC,” DOI: [10.6084/m9.figshare.31211437](https://doi.org/10.6084/m9.figshare.31211437) (2026).
- [4] M. Magnusson, “EFC Closure Conjectures,” DOI: [10.6084/m9.figshare.31224466](https://doi.org/10.6084/m9.figshare.31224466) (2026).
- [5] S. S. McGaugh, F. Lelli, J. M. Schombert, “Radial Acceleration Relation in Rotationally Supported Galaxies,” *Phys. Rev. Lett.* **117**, 201101 (2016).